

B.Tech. Degree III Semester Examination November 2013

CS 1302 LOGIC DESIGN (2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 x 5 = 40)

- I. (a) Convert (i) $(63.25)_{10}$ to Hexadecimal and octal.
(ii) $(117)_{16}$ to BCD and Excess-3 code
- (b) State De-Morgan's theorem. Show that a 2-level AND – OR network is equivalent to 2-level NAND-NAND network.
- (c) Explain a 2-to-4 line decoder. Implement a full adder using ONE 3 input decoder and gates.
- (d) What is a sequential circuit? Explain a JK flipflop with neat circuit diagram and obtain its characteristic table.
- (e) Explain a 4 bit ring counter. Obtain its counting sequence.
- (f) Obtain the hamming code for the message $(1101)_2$ using odd parity.
- (g) What are hazards? Explain static-1, static-0 and dynamic hazards.
- (h) Explain TTL NAND gate with neat diagram.

PART B

(4 x 15 = 60)

- II. (a) What are gray codes? What are its properties? Find the gray code next to the code "10110110" in an 8 bit gray code sequence. (8)
- (b) Simplify; $F_{(A,B,C,D,E)} = \sum(0, 2, 4, 6, 9, 13, 21, 23, 24, 25, 26, 29, 31)$ that has dont care conditions. (7)
 $d_{(A,B,C,D,E)} = \sum(8, 10, 16, 18)$, using k-map method. Implement the circuit using NAND gates only.
- OR**
- III. (a) What are universal gates? With necessary expressions implement NOT, AND, OR and XOR using NAND gates only. (10)
- (b) Using Boolean laws, show that $AB + \bar{A}C + BC = AB + \bar{A}C$ (5)
- IV. Discuss the disadvantages of parallel adder. With necessary expressions, design a 4-bit "Carry Look Ahead adder" using combinational circuits. (15)

OR

(P.T.O.)

- V. (a) Design a 4:1 multiplexer using gates. (5)
 (b) Implement the given Pos $Y = \pi(0, 2, 4, 7)$ on a 4:1 MUX. (5)
 (c) Reduce the number of states in the following state table and write the reduced table. (5)

Present state	Next State		Output	
	X = 0	X = 1	X = 0	X = 1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	g	f	0	1
g	a	f	0	1

- VI. (a) Explain a 4 bit SISO shift register with neat diagram. (5)
 (b) Draw the logic diagram of a RAM cell. Explain the control lines. (5)
 (c) Draw the logical construction of a 4×3 RAM module. (5)
- OR**
- VII. (a) Implement a BCD Ripple Counter using JK flipflop and AND gates. Draw its state diagram. (8)
 (b) Give the block diagram and explain $((n \times k \times m)$ PLA. Compare PLA and PAL. (7)
- VIII. (a) Explain a "key board – de bounce" circuit using SR latch. (5)
 (b) Write short notes on: (10)
 (i) Fan out (ii) Propagation delay (iii) Noise Margin
- OR**
- IX. (a) Draw circuit and explain: (10)
 (i) DTL NAND gate
 (ii) CMOS inverter
 (b) Compare TTL and CMOS logic families. (5)
